

Introduction to Artificial Intelligence and Machine Learning

Core Concepts of AI Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn. AI spans subfields including Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Computer Vision.

Machine Learning Paradigms Machine Learning is divided into three primary paradigms: 1. **Supervised Learning:** The model learns from labeled training data (inputs and corresponding targets), such as linear regression, decision trees, and support vector machines. 2. **Unsupervised Learning:** The model finds hidden patterns or intrinsic structures in input data without labels. Examples include K-Means clustering, Principal Component Analysis (PCA), and autoencoders. 3. **Reinforcement Learning:** An agent learns to make decisions by performing actions in an environment to maximize cumulative reward.

Deep Learning and Neural Networks Deep Learning utilizes multi-layered artificial neural networks (ANNs) inspired by the human brain. Key architectures include Convolutional Neural Networks (CNNs) for spatial and image data, and Recurrent Neural Networks (RNNs) for sequential data.